

MODULE- 02

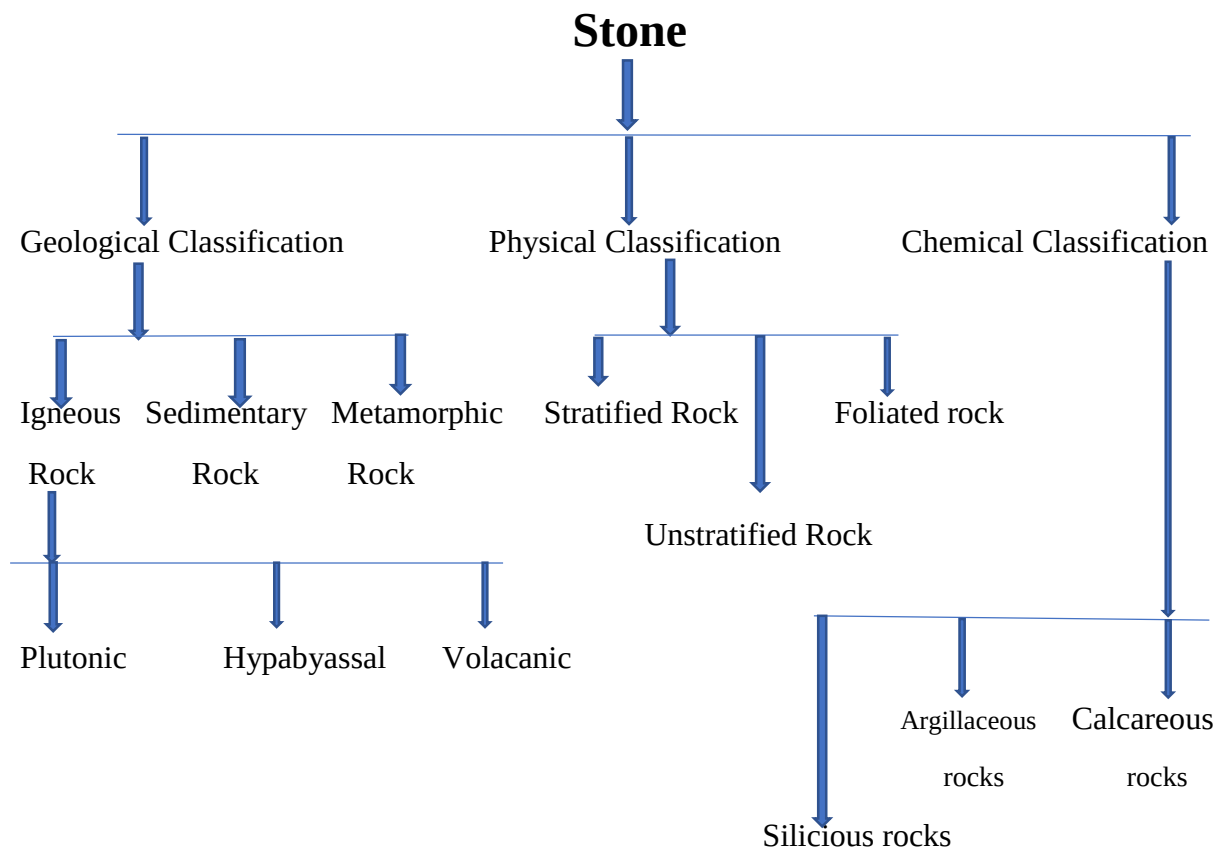
STONE

Stone is a ‘naturally available building material’ which has been used from the early age of civilization. It is available in the form of rocks, which is cut to required size and shape and used as building block. It has been used to construct small residential buildings to large palaces and temples all over the world.

Type of Stones

Stones used for civil engineering works may be classified in the following three ways:

1. Geological Classification
2. Physical Classification
3. Chemical Classification



Igneous Rock

The rocks which are formed by the cooling of magma are called Igneous Rock.

It is of 2 types

1. Intrusive Rock

2. Extrusive Rock

Intrusive Rock: When the magma comes below the earth surface and solidify then the rock is known as Intrusive rock.

Extrusive Rock: When the magma comes above the earth surface and solidify, then the rock is known as Extrusive Rock.

The igneous rocks are recognized in the following 3 classes

1. Plutonic

2. Hypabyssal

3. Volcanic

Plutonic

1. Cooling of magma at a considerably depth from the earth surface.

2. The process is slow and rock possess coarsely grained crystalline structure.

EX-Granite

Hypabyssal

1. Cooling of magma at a shallow depth from the earth surface.

2. The process is quick and rock possess finely grained crystalline structure.

EX-Dolerite

Volcanic

1. These are formed due to the pouring of Magma at earth's surface.

2. The process is very rapid and rock possess extremely fine Crystalline structure.

EX-Basalt

Sedimentary Rock

These rocks are formed by the deposition of products of weathering on the pre-existing rocks.

All the products of weathering are carried away from the place of origin by the agents of transport like Frost, rain, wind and flowing water.

Example-Gravel, Lime stone, Sand stone, Gypsum, Siltstone, Shale, Kankar, Chalk

Metamorphic Rock

These rocks are formed by the change in character of Pre-existing rocks. These rocks are change their character when they are subjected to great heat and pressure. The pressure change is known as Metamorphism.

There are 3 agents of metamorphism that is heat, pressure and chemically acting fluids.

Ex-Quartzite, Marble, Slate, Laterite

Physical Classification

This classification is based on the structure of rocks.

According to this Rocks are classified into 3 types

- 1.Stratified Rock
- 2.Unstratified Rock
- 3.Foliated Rock

1.Stratified Rocks: These rocks are having layered structure. They possess planes of stratification or cleavage. They can be easily split along these planes. Sand stones, lime stones, slate etc. are the examples of this class of stones.

2.Unstratified Rocks: These rocks are not stratified. They possess crystalline and compact grains. They cannot be split in to thin slab. Granite, trap, marble etc. are the examples of this type of rocks.

3. Foliated Rocks: These rocks have a tendency to split along a definite direction only. The direction need not be parallel to each other as in case of stratified rocks. This type of structure is very common in case of metamorphic rocks.

Chemical Classification

According to this, Rocks are classified into 3 types

1. Silicious Rock
2. Argillaceous Rock
3. Calcareous rocks

Silicious rocks: The main content of these rocks is silica. They are hard and durable. Examples of such rocks are granite, trap, sand stones etc.

Argillaceous rocks: The main constituent of these rocks is argil *i.e.*, clay. These stones are hard and durable but they are brittle. They cannot withstand shock. Slates and laterites are examples of this type of rocks.

Calcareous rocks: The main constituent of these rocks is calcium carbonate. Limestone is a calcareous rock of sedimentary origin while marble is a calcareous rock of metamorphic origin.

Properties of Stones

The following properties of the stones should be looked into before selecting them for engineering works:

Structure: The structure of the stone may be stratified (layered) or unstratified. Structured stones should be easily dressed and suitable for super structure. Unstratified stones are hard and difficult to dress. They are preferred for the foundation works.

Texture: Fine grained stones with homogeneous distribution look attractive and hence they are used for carving. Such stones are usually strong and durable.

Density: Denser stones are stronger. Light weight stones are weak. Hence stones with specific gravity less than 2.4 are considered unsuitable for buildings.

Appearance: A stone with uniform and attractive colour is durable, if grains are compact. Marble and granite get very good appearance, when polished. Hence they are used for face works in Buildings.

Strength: Strength is an important property to be looked into before selecting stone as building block. Indian standard code recommends, a minimum crushing strength of 3.5 N/mm² for any building block.

Porosity and Absorption: All stones have pores and hence absorb water. The reaction of water with material of stone cause disintegration. Absorption test is specified as percentage of water by the stone when it is immersed under water for 24 hours. For a good stone it should be as small as possible and in no case more than 5.

Weathering: Rain and wind cause loss of good appearance of stones. Hence stones with good weather resistance should be used for face works.

Toughness: The resistance to impact is called toughness. It is determined by impact test. Stones with toughness index more than 19 are preferred for road works. Toughness index 13 to 19 are considered as medium tough and stones with toughness index less than 13 are poor stones.

Resistance to Fire: Sand stones resist fire better. Argillaceous materials, though poor in strength, are good in resisting fire.

Ease in Dressing: Cost of dressing contributes to cost of stone masonry to a great extent. Dressing is easy in stones with lesser strength. Hence an engineer should look into sufficient strength rather than high strength while selecting stones for building works.

Seasoning: The stones obtained from quarry contain moisture in the pores. The strength of the stone improves if this moisture is removed before using the stone. The process of removing moisture from pores is called seasoning. The best way of seasoning is to allow it to the action of nature for 6 to 12 months. This is very much required in the case of laterite stones.

Requirements of Good Building Stones

The following are the requirements of good building stones:

Strength: The stone should be able to resist the load coming on it. Ordinarily this is not of primary concern since all stones are having good strength. However in case of large structure, it may be necessary to check the strength.

Durability: Stones selected should be capable of resisting adverse effects of natural forces like wind, rain and heat.

Hardness: The stone used in floors and pavements should be able to resist abrasive forces caused by movement of men and materials over them.

Toughness: Building stones should be tough enough to sustain stresses developed due to vibrations. The vibrations may be due to the machinery mounted over them or due to the loads moving over them. The stone aggregates used in the road constructions should be tough.

Specific Gravity: Heavier variety of stones should be used for the construction of dams, retaining walls, docks and harbours. The specific gravity of good building stone is between 2.4 and 2.8.

Porosity and Absorption: Building stone should not be porous. If it is porous rain water enters into the pore and reacts with stone and crumbles it. In higher altitudes, the freezing of water in pores takes place and it results into the disintegration of the stone.

Dressing: Giving required shape to the stone is called dressing. It should be easy to dress so that the cost of dressing is reduced. However the care should be taken so that, this is not be at the cost of the required strength and the durability.

Appearance: In case of the stones to be used for face works, where appearance is a primary requirement, its colour and ability to receive polish is an important factor.

Seasoning: Good stones should be free from the quarry sap. Laterite stones should not be used for 6 to 12 months after quarrying. They are allowed to get rid of quarry sap by the action of nature. This process of removing quarry sap is called seasoning.

Uses of Stones

- (i) Stone masonry is used for the construction of foundations, walls, columns and arches.
- (ii) Stones are used for flooring.
- (iii) Stones with good appearance are used for the face works of buildings. Polished marbles and granite are commonly used for face works.
- (iv) Stones are used for paving of roads, footpaths and open spaces round the buildings.
- (v) Stones are also used in the constructions of piers and abutments of bridges, dams and retaining walls.
- (vi) Crushed stones with gravel are used to provide base course for roads. When mixed with tar they form finishing coat.
- (vii) Crushed stones are used in the following works also:
 - (a) As a basic inert material in concrete
 - (b) For making artificial stones and building blocks
 - (c) As railway ballast.

BRICK

A rectangular block typically made of fired or sun-dried clay used in construction is called as Brick.

There are 2 faces of a Brick

1. Stretcher Brick (the side face brick)
2. Header Brick (the end surface of brick)

Composition of Brick

S- Silica (50-60%)

A-Alumina (20-30%)

L-Lime (4-5%)

O-Iron Oxide (5-6%)

M-Magnesia (1%)

Silica

- Prevents cracking, shrinkage, warping of bricks.
- Gives a uniform shape to the brick
- Decrease shrinkage during burning.

- If present excess amount, then destroys the corrosion between the particles, hence brick become brittle.

Alumina

- Imparts plasticity to the brick, so that brick can be moulded to any desired shape.
- If present in excess quantity, then causes shrinkage and warping during and makes the brick too hard during working process.

Lime

- Also prevents shrinkage of brick.
- Helps silica in clay to melt and helps in binding but if it in excess causes brick to melt during the burning thereby it results in loss of shape of brick.

Oxide of Iron

- Develops strength of brick.
- Excess quantity makes brick blue colour.
- Imparts reddish brown colour
- Improves impermeability.

Magnesia

- Prevents shrinkage and imparts yellowish tint color.

Classification of Bricks Based on their Quality

The bricks used in construction are classified as:

(i) First class bricks

(ii) Second class bricks

(iii) Third class bricks and

(iv) Fourth class bricks

(i) **First Class Bricks:** These bricks are of standard shape and size. They are burnt in kilns.

They fulfill all desirable properties of bricks.

(ii) **Second Class Bricks:** These bricks are ground moulded and burnt in kilns. The edges may not be sharp and uniform. The surface may be somewhat rough. Such bricks are commonly used for the construction of walls which are going to be plastered.

(iii) **Third Class Bricks:** These bricks are ground moulded and burnt in clamps. Their edges are somewhat distorted. They produce dull sound when struck together. They are used for temporary and unimportant structures.

(iv) **Fourth Class Bricks:** These are the over burnt bricks. They are dark in colour. The shape is irregular. They are used as aggregates for concrete in foundations, floors and roads.

Properties of Bricks

The following are the required properties of good bricks:

- (i) **Colour:** Colour should be uniform and bright.
- (ii) **Shape:** Bricks should have plane faces. They should have sharp and true right angled corners.
- (iii) **Size:** Bricks should be of standard sizes as prescribed by codes.
- (iv) **Texture:** They should possess fine, dense and uniform texture.
- (v) **Soundness:** When struck with hammer or with another brick, it should produce metallic sound.
- (vi) **Hardness:** Finger scratching should not produce any impression on the brick.
- (vii) **Strength:** Crushing strength of brick should not be less than 3.5 N/mm². A field test for strength is that when dropped from a height of 0.9 m to 1.0 m on a hard ground, the brick should not break into pieces.
- (viii) **Water Absorption:** After immersing the brick in water for 24 hours, water absorption should not be more than 20 per cent by weight. For class-I works this limit is 15 per cent.
- (ix) **Efflorescence:** Bricks should not show white patches when soaked in water for 24 hours and then allowed to dry in shade. White patches are due to the presence of sulphate of calcium, magnesium and potassium. They keep the masonry permanently in damp and wet conditions.
- (x) **Thermal Conductivity:** Bricks should have low thermal conductivity, so that buildings built with them are cool in summer and warm in winter.
- (xi) **Sound Insulation:** Heavier bricks are poor insulators of sound while light weight and hollow bricks provide good sound insulation.
- (xii) **Fire Resistance:** Fire resistance of bricks is usually good. In fact bricks are used to encase steel columns to protect them from fire.

Uses of Bricks

- (i) As building blocks.
- (ii) For protecting steel columns from fire.
- (iii) As aggregates in providing water proofing to R.C.C. roofs.
- (iv) For pavers for footpaths and cycle tracks.
- (v) For lining sewer lines.

Qualities of a good Brick

The good bricks which are to be used for the construction of important structures should possess the following qualities.

- The bricks should be uniform in shape and should be of standard size.
- The brick should give a clear metallic ringing sound when stuck with each other.
- The brick when broken or fractured should possess a bright homogeneous and uniform compact structure free from voids.
- The brick should not absorb water more than 20% by weight for first class brick and 22% by weight for second class brick when worked in cold water for a period of 24 hours.
- The brick should be sufficiently hard. No impression should be left on the brick surface when it is scratched with finger nail.
- The bricks should not break into pieces when dropped flat on hard ground from a height of about one metre.
- The bricks when soaked into water for 24 hours , should not show deposits of white salts when allowed to dry in shade.
- No brick should have crushing strength below 5.50 N/MM^2

Size of Brick

Standard Size- $19\text{cm} \times 9\text{cm} \times 9\text{cm}$ or $190\text{mm} \times 90\text{mm} \times 90\text{mm}$ (Without mortar)

Nominal Size- $20\text{cm} \times 10\text{cm} \times 10\text{cm}$ or $200\text{mm} \times 100\text{mm} \times 100\text{mm}$ (With mortar)

CEMENT

Cement is a commonly used binding material in the construction. The cement is obtained by burning a mixture of calcareous (calcium) and argillaceous (clay) material at a very high temperature and then grinding the clinker so produced to a fine powder.

Composition of Cement

L-Lime	M-Magnesia
A-Alumina	S-Sulphur
S-Silica	A-Alkalies
G-Gypsum	
I-Iron oxide	

Lime (62-65%)

- Imparts strength and soundness
- If it is in excess amount then it difficult to combine with other compound.
- Deficiency reduces strength of cement, Set quickly.

Alumina (3-8%)

- Imparts quick setting property.
- Excess amount weakness the cement.

Silica (17-25%)

- Imparts strength of cement
- Excess amount increases the setting time.

Iron-oxide (3-4%)

- Imparts colour, strength, hardness to the cement.
- Induces reddish brown tint to the cement.

Magnesia (1-3%)

- Imparts strength, colour to the cement.
- Induces yellow tint colour to the cement.

Sulphur (1-3%)

- Responsible for imparting soundness.

Alkalis (0.2-1%)

- Causes efflorescence.
- It reacts with water and form white grey spots.
- Also accelerates setting time of paste.

Properties of Ordinary Portland Cement

Chemical properties: Portland cement consists of the following chemical compounds:

(a) Tri calcium silicate	3 CaO.SiO ₂ (C ₃ S)	40%
(b) Di calcium silicate	2CaO.SiO ₂ (C ₂ S)	30%
(c) Tri calcium aluminate	3CaO.Al ₂ O ₃ (C ₃ A)	11%
(d) Tetra calcium aluminate	4CaO.Al ₂ O ₃ .Fe ₂ O ₃ (C ₃ AF)	11%

When water is added to cement, C3A is the first to react and cause initial set. It generates great amount of heat. C3S hydrates early and develops strength in the first 28 days. It also generates heat. C2S is the next to hydrate. It hydrates slowly and is responsible for increase in ultimate strength. C4AF is comparatively inactive compound.

Physical properties:

- a. Fineness
- b. Soundness
- c. Setting Time
- d. Crushing Strength

Fineness

It is measured in terms of percentage of weight retained after sieving the cement through 90-micron sieve or by surface area of cement in square centimetres per gramme of cement. According to IS code specification weight retained on the sieve should not be more than 10 per cent. In terms of specific surface should not be less than 2250 cm²/gm.

Soundness

Once the concrete has hardened it is necessary to ensure that no volumetric changes takes place. The cement is said to be unsound, if it exhibits volumetric instability after hardening. IS code recommends test with Le Chatelier mould for testing this property. At the end of the test, the indicator of Le Chatelier mould should not expand by more than 10 mm.

Setting Time

A period of 30 minutes as minimum setting time for initial setting and a maximum period of 600 minutes as maximum setting time is specified by IS code, provided the tests are conducted as per the procedure prescribed by IS 269-1967.

Crushing Strength

For this mortar cubes are made with standard sand and tested in compression testing machine as per the specification of IS code. The minimum strength specified is 16 N/mm² after 3 days and 22 N/mm² after 7 days of curing.

Uses of Cement

Cement is used widely for the construction of various structures. Some of them are listed below:

- (i) Cement slurry is used for filling cracks in concrete structures.
- (ii) Cement mortar is used for masonry work, plastering and pointing.
- (iii) Cement concrete is used for the construction of various structures like buildings, bridges, water tanks, tunnels, docks, etc.
- (iv) Cement is used to manufacture lamp posts, telephone posts, railway sleepers, piles etc.
- (v) For manufacturing cement pipes, garden seats, dust bins, flower pots etc. cement is commonly used.
- (vi) It is useful for the construction of roads, footpaths, courts for various sports etc.

***TYPES OF CEMENT –ASSIGNMENT**

AGGREGATE

These are the inert or chemically inactive materials which form the bulk of cement concrete.

These aggregates are bound together by means of cement. These aggregates are classified into 2 types

1. Fine aggregates

2. Coarse aggregates

The material which is passed through BIS test sieve no.480 is termed as a fine aggregate.

Usually the natural river sand is used as a fine aggregate. But at places, where natural sand is not available economically the finely crushed stone may be used as a fine aggregate.

The material which is retained on BIS test sieve no.480 is termed as coarse aggregate. The broken stone is generally used as a coarse aggregate.

The aggregates to be used for cement concrete work should be hard, durable, and clean. The aggregate should be completely free from lumps of clay, Organic and vegetable matter, fine dust etc. The presence of all such debris prevents adhesion of aggregates and hence reduces the strength of concrete.

The aggregate may be classified into 2 categories

- (i) Natural aggregate
- (ii) Artificial aggregates

Requirements of a Good Aggregate

Following are the desirable properties or requirements of a good aggregate.

- | | |
|---------------|--------------|
| 1. Adhesion | 4. Shape |
| 2. Durability | 5. Strength |
| 3. Hardness | 6. Toughness |

Adhesion

The aggregates which are to be used for the construction should have less affinity with water as compared with the binding material. If this quality is absent in the aggregate, it will lead to the separation of bituminous or cement coating in the presence of water.

Durability

The durability of an aggregate indicates its resistance to the action of weather. It is desirable that the aggregate should possess sufficient soundness to resist the action of weather and age so that the life of the structure made with it may be prolonged.

Hardness

The aggregates should be reasonably hard to offer resistance to the action of abrasion and attrition. The aggregates are always subjected to constant rubbing action. It is known as abrasion. The mutual rubbing of stone is known as attrition and it may cause little wear in the aggregates.

Shape

The shape of aggregates may be rounded, cubical, angular, flaky or elongated. The flaky and elongated particles possess less strength and durability and their use in the construction should be avoided as far as possible. The rounded particles are preferred in cement concrete construction.

Strength

The aggregates should be sufficiently strong to withstand the stress developed due to the wheel loads of the traffic.

Toughness

The toughness of an aggregate is that property which enables the aggregate to resist the fracture when struck with a hammer and it is necessary in a metal to withstand the impact blows by the traffic. It is desirable that the aggregate is reasonably tough.

MORTAR

The term mortar is used to indicate a paste prepared by adding required quantity of water to a mixture of binding material like cement or lime and fine aggregates like sand. The above 2 components of mortar namely the binding material and fine aggregates are sometimes called as matrix and adulterant respectively.

Properties of Good Mortar

Following are the properties of a good mortar

- 1.It should be capable of developing good adhesion with the building units such as bricks, stones, etc.
- 2.It should be capable of resisting penetration of rain water.
- 3.It should be cheap.
- 4.It should be durable.
- 5.It should be workable.
- 6.It should not affect the durability of materials with which it comes into contact.
- 7.It should set quickly so that speed in construction may be achieved.
- 8.The joints formed by mortar should not develop cracks and they should be able to maintain their appearance for a sufficiently long period.

Uses of Mortar

Following are the uses of mortar

- 1.To bind the building units such as bricks, stones etc into a solid mass.
- 2.to carry out pointing and plaster work on exposed surface of masonry.
- 3.to form joints in the pipes.
- 4.to improve general appearance of structure.
- 5.To hide the open joints of brickwork and stone work.
- 6.to fill up the cracks detected in the structure during the maintenance process etc.

***DIFFERENT TYPES OF MORTAR AND THEIR PROPERTIES- ASSIGNMENT**

CONCRETE

Plain concrete, commonly known as concrete, is an intimate mixture of binding material, fine aggregate, coarse aggregate and water. This can be easily moulded to desired shape and size before it loses plasticity and hardens.

Proportioning of Concrete

- The process of selection of relative proportions of cement, sand, coarse aggregate and water so as to obtain a concrete of desired quality is known as the proportioning of concrete.
- The concrete as per IS:456:2000 is designated in 7 grades, namely M10, M15, M20, M25, M30, M35, M40. The letter M refers to the mix and the number refers to the specified compressive strength of that mix at 28 days expressed in N/mm².
- For general guidance, the nominal mixes correspond approximately to the different grades as follows.

M5-1:5:10

M7.5-1:4:8

M10-1:3:6

M15-1:2:4

M20-1:1.5:3

M25-1:1:2

- All the above proportions are referred to as the ratio of weights of cement to sand to coarse aggregate.

For example, in 1:5:10

1 refers to the weight of cement

5 refers to the weight of sand

10 refers to the weight of coarse aggregate

- In proportioning of concrete it is kept in mind that voids in coarse aggregates are filled with sand and the voids in sand are filled with cement paste.

Water-Cement Ratio

- The ratio of amount of water to the amount of cement by weight is termed as water cement ratio and the strength and quality of concrete is primarily dependent upon this ratio.
- The quantity of water is usually expressed in litres per bag of cement and hence the water-cement ratio reduces the quantity of water required in litres per kg of cement as 1 litre of water weighs 1 kg. For instance, if water required for 1 bag of cement (50kg) is 30 litres, the water cement ratio is $30/50 = .60$
- Strength of concrete is inversely proportional to the water-cement ratio.

- The water-cement ratio for structures which are exposed to weather should be carefully decided. For instances for structures which are regularly wetting and drying, the water cement ratio by weight should be .45 and .55 for thin sections and mass respectively. For structures which are continuously under water, the water-cement ratio by weight should be .55 and .65 for thin sections and mass concrete respectively.

Preparing and Placing of Concrete

The following steps are involved in the concreting:

1. Batching
2. Mixing
3. Transporting and placing
4. Compacting

1. **Batching:** The measurement of materials for making concrete is known as batching. The following two methods of batching is practiced:

(a) Volume batching

(b) Weight batching.

(a) **Volume Batching:** In this method cement, sand and concrete are batched by volume. A gauge box is made with wooden plates, its volume being equal to that of one bag of cement. One bag of cement has volume of 35 litres. The required amount of sand and coarse aggregate is added by measuring on to the gauge box. The quantity of water required for making concrete is found after deciding water cement ratio. For example, if water cement ratio is 0.5, for one bag of cement (50 kg), water required is $0.5 \times 50 = 25$ kg, which is equal to 25 litres. Suitable measure is used to select required quantity of water. Volume batching is not ideal method of batching. Wet sand has higher volume for the same weight of dry sand. It is called bulking of sand. Hence it upsets the calculated volume required.

(b) **Weight Batching:** This is the recommended method of batching. A weighing platform is used in the field to pick up correct proportion of sand and coarse aggregates. Large weigh batching plants have automatic weighing equipment's.

2. **Mixing:** To produce uniform and good concrete, it is necessary to mix cement, sand and coarse aggregate, first in dry condition and then in wet condition after adding water.

The following methods are practiced:

(a) Hand Mixing

(b) Machine Mixing.

(a) **Hand Mixing:** Required amount of coarse aggregate for a batch is weighed and is spread on an impervious platform. Then the sand required for the batch is spread over coarse

aggregate. They are mixed in dry condition by overturning the mix with shovels. Then the cement required for the batch is spread over the dry mix and mixed by shovels. After uniform texture is observed water is added gradually and mixing is continued. Full amount of water is added and mixing is completed when uniform colour and consistency is observed. The process of mixing is completed in 6–8 minutes of adding water. This method of mixing is not very good but for small works it is commonly adopted.

(b) **Machine Mixing:** In large and important works machine mixing is preferred. Required quantities of sand and coarse aggregates are placed in the drum of the mixer. 4 to 5 rotations are made for dry mixing and then required quantity of cement is added and dry mixing is made with another 4 to 5 rotations. Water is gradually added and drum is rotated for 2 to 3 minutes during which period it makes about 50 rotations. At this stage uniform and homogeneous mix is obtained.

3. Transporting and Placing of Concrete. After mixing concrete should be transported to the final position. In small works it is transported in iron pans from hand to hand of a set of workers. Wheel barrow and hand carts also may be employed. In large scale concreting chutes and belt conveyors or pipes with pumps are employed. In transporting care should be taken to see that segregation of aggregate from matrix of cement do not take place.

Concrete is placed on form works. The form works should be cleaned and properly oiled. If concrete is to be placed for foundation, the soil bed should be compacted well and is made free from loose soil.

Concrete should be dropped on its final position as closely as possible. If it is dropped from a height, the coarse aggregates fall early and then mortar matrix. This segregation results into weaker concrete.

4. Compaction of Concrete: In the process of placing concrete, air is entrapped. The entrapped air reduces the strength of concrete up to 30%. Hence it is necessary to remove this entrapped air. This is achieved by compacting the concrete after placing it in its final position. Compaction can be carried out either by hand or with the help of vibrators.

(a) **Hand Compaction:** In this method concrete is compacted by ramming, tamping, spading or by slicing with tools. In intricate portions a pointed steel rod of 16 mm diameter and about a metre long is used for poking the concrete.

(b) **Compaction by Vibrators:** Concrete can be compacted by using high frequency vibrators. Vibration reduces the friction between the particles and sets the motion of particles. As a result entrapped air is removed and the concrete is compacted. The use of vibrators reduces the compaction time. When vibrators are used for compaction, water cement ratio can be less,

which also help in improving the strength of concrete. Vibration should be stopped as soon as cement paste is seen on the surface of concrete. Over vibration is not good for the concrete.

The following types of vibrators are commonly used in concreting:

- (a) Needle or immersion vibrators
- (b) Surface vibrators
- (c) Form or shutter vibrators
- (d) Vibrating tables.

Needle vibrators are used in concreting beams and columns. Surface vibrators and form vibrators are useful in concreting slabs. Vibrating tables are useful in preparing precast concrete elements.

Curing of Concrete

Curing may be defined as the process of maintaining satisfactory moisture and temperature conditions for freshly placed concrete for some specified time for proper hardening of concrete. Curing in the early ages of concrete is more important. Curing for 14 days is very important. Better to continue it for 7 to 14 days more. If curing is not done properly, the strength of concrete reduces. Cracks develop due shrinkage. The durability of concrete structure reduces. The following curing methods are employed:

- (a) Spraying of water
- (b) Covering the surface with wet gunny bags, straw etc.
- (c) Ponding
- (d) Steam curing and
- (e) Application of curing compounds.

(a) **Spraying of water:** Walls, columns, plastered surfaces are cured by sprinkling water.

(b) **Wet covering the surface:** Columns and other vertical surfaces may be cured by covering the surfaces with wet gunny bags or straw.

(c) **Ponding:** The horizontal surfaces like slab and floors are cured by stagnating the water to a height of 25 to 50 mm by providing temporary small hunds with mortar.

(d) **Steam curing:** In the manufacture of pre-fabricated concrete units steam is passed over the units kept in closed chambers. It accelerates curing process, resulting into the reduction of curing period.

(e) **Application of curing compounds:** Compounds like calcium chloride may be applied on the curing surface. The compound shows affinity to the moisture and retains it on the surface. It keeps the concrete surface wet for a long time.

Properties of Concrete

- It has a compressive strength
- It is free from corrosion and there is no appreciable effect of atmospheric agents on it.
- It hardens with age and the process of hardening continues for a long time after the concrete has attained sufficient strength.
- It has a tendency to shrink. There is a initial shrinkage of cement concrete which is mainly due to the loss of water through of forms, absorptions by surface of forms. The shrinkage of concrete occurs as it hardens. This tendency of cement concrete can be minimized by proper curing of concrete.
- It has a tendency to be porous. This is due to the presence of voids which is formed during and after its placing.
- It forms a hard surface, capable of resisting abrasion.

Uses of Concrete

1. As bed concrete below column footings, wall footings, on wall at supports to beams
2. As sill concrete
3. Over the parapet walls as coping concrete
4. For flagging the area around buildings
5. For pavements
6. For making building blocks.

TIMBER

Timber refers to wood used for construction works. Timber was used as building material even by primitive man. Many ancient temples, palaces and bridges built with timber can be seen even today.

Classification of Timber

Various bases are considered for the classification of timbers. The following are the important basis:

- (i) Mode of growth
- (ii) Modulus of elasticity
- (iii) Durability
- (iv) Grading
- (v) Availability.

(i) Classification Based on Mode of Growth: On the basis of mode of growth trees are classified

as (a) Exogeneous and (b) Endogeneous

(a) **Exogeneous Trees:** These trees grow outward by adding distinct consecutive ring every year. These rings are known as annual rings. Hence it is possible to find the age of timber by counting these annual rings. These trees may be further divided into (1) coniferous and (2) deciduous.

Coniferous trees are having cone shaped leaves and fruits. The leaves do not fall till new ones are grown. They yield soft wood.

Deciduous trees are having broad leaves. These leaves fall in autumn and new ones appear in springs. They yield strong wood and hence they are commonly used in building construction.

(b) **Endogeneous Trees:** These trees grow inwards. Fresh fibrous mass is in the inner most portion. Examples of endogenous trees are bamboo and cane. They are not useful for structural works.

(ii) **Classification Based on Modulus of Elasticity:** Young's modulus is determined by conducting bending test. On this basis timber is classified as:

Group A: $E = 12.5 \text{ kN/mm}^2$ (E-Modulus of elasticity)

Group B: $E = 9.8 \text{ kN/mm}^2$ to 12.5 kN/mm^2

Group C: $E = 5.6 \text{ kN/mm}^2$ to 9.8 kN/mm^2 .

(iii) **Classification Based on Durability:** Durability tests are conducted by the forest research establishment. They bury test specimen of size $600 \times 50 \times 50 \text{ mm}$ in the ground to half their length and observe their conditions regularly over several years. Then timbers are classified as:

High durability: If average life is more than 10 years.

Moderate durability: Average life between 5 to 10 years.

Low durability: Average life less than 5 years.

(iv) **Classification Based on Grading:** IS 883-1970 classifies the structural timber into three grades-select grade, grade I and grade II. The classification is based on permissible stresses, defects etc.

(v) **Classification Based on Availability:** Forest departments classify timbers based on the availability as

X—Most common. 1415 m^3 or more per year

Y—Common. 355 m^3 to 1415 m^3 per year

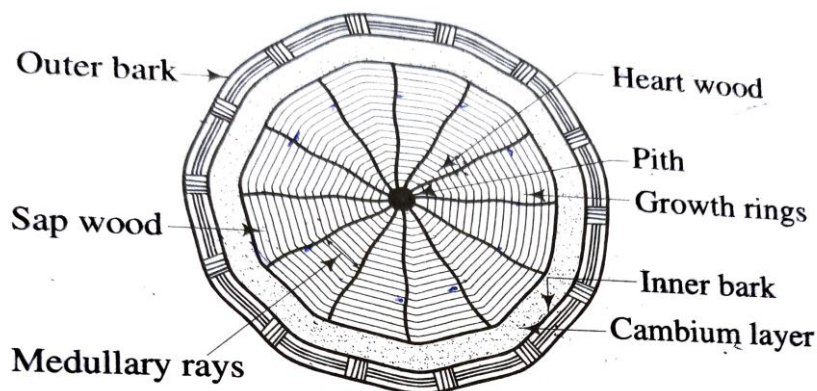
Z—Less common. Less than 355 m^3 per year.

Cross-section of Exogenous Tree

1. **Pith:** It is the inner most part of the tree and hence the oldest part of exogenous tree when the plant becomes old, the pith dies and becomes fibrous and dark. It varies in size and shape.
2. **Heart Wood:** This is the portion surrounding pith. It is dark in colour and strong. This portion is useful for various engineering purpose. This is the dead part of wood. It consists of several annular rings.
3. **Sap Wood:** It is the layer next to heart wood. It denotes recent growth and contains sap. It takes active part in the growth of trees by allowing sap to move in upward direction. The annual rings of sap wood are less sharply divided and are light in colour.
4. **Cambium Layer:** It is a thin layer of fresh sap lying between sap wood and the inner bark. It contains sap which is not yet converted into sap wood. If the bark is removed and cambium layer is exposed to atmosphere, cells cease to be active and tree dies.
5. **Inner Bark:** It is a inner skin of tree protecting the cambium layer. It gives protection to cambium layer.
6. **Outer Bark:** It is the outer skin of the tree and consists of wood fibres. Sometimes it contains fissures and cracks.
7. **Medullary Rays:** These are thin radial fibres extending from pith to cambium layer. They hold annular rings together. In some of trees they are broken and some other they may not be prominent.

Timber

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Cross-section of an exogenous tree

FIG. 9-1

α but merely a prolongation of the pith of stem

Properties of Timber

Properties of good timbers are:

Colour: It should be uniform.

Odour: It should be pleasant when cut freshly.

Soundness: A clear ringing sound when struck indicates the timber is good.

Texture: Texture of good timber is fine and even.

Grains: In good timber grains are close.

Density: Higher the density stronger is the timber.

Hardness: Harder timbers are strong and durable.

Warping: Good timber do not warp under changing environmental conditions.

Toughness: Timber should be capable of resisting shock loads.

Abrasion: Good timber do not deteriorate due to wear. This property should be looked into, if timber is to be used for flooring.

Strength: Timber should have high strength in bending, shear and direct compression.

Fire resistance: A good timber should have high resistance to fire.

Permeability: Good timber has low water permeability.

Workability: Timber should be easily workable. It should not clog the saw.

Durability: Good timber is one which is capable of resisting the action of fungi and insects attack

Defects: Good timber is free from defects like dead knots, shakes and cracks.

Uses of Timber

Timber is used for the following works:

1. For heavy construction works like columns, trusses, piles.
2. For light construction works like doors, windows, flooring and roofing.
3. For other permanent works like for railway sleepers, fencing poles, electric poles and gates.
4. For temporary works in construction like scaffolding, centering, shoring and strutting, packing of materials.
5. For decorative works like showcases and furniture.
6. For body works of buses, lorries, trains and boats
7. For industrial uses like pulps (used in making papers), card boards, wall papers
8. For making sports goods and musical instruments.

PAINTS

Paints are applied on the surfaces of timber, metals and plastered surfaces as a protective layer and at the same time to get pleasant appearance. Paints are applied in liquid form and after sometime the volatile constituent evaporates and hardened coating acts as a protective layer.

Properties of an Ideal Paint

1. It should be possible to apply easily and freely.
2. It should dry in reasonable time.
3. It should form hard and durable surface.
4. It should not be harmful to the health of workers.
5. It should not be easily affected by atmosphere.
6. It should possess attractive and pleasing appearance.
7. It should form a thin film of uniform nature *i.e*, it should not crack.
8. It should possess good spreading power.
9. It should be cheap.

PLASTICS

Plastic is an organic material prepared out of resin. It may or may not contain fillers, plasticisers and solvents. Plastic may be defined as a natural or synthetic organic material which are having the property of being plastic at some stage of their manufacture when they can be moulded to required size and shape.

Types of Plastics

Primarily there are two types of plastics:

1. Thermosetting and
2. Thermoplastic.

1. **Thermosetting Plastics:** It needs momentary heated condition and great pressure during shaping. When heated cross linkage is established between the molecules and chemical reaction takes place. During this stage shape can be changed with pressure. This change is not reversible. The scrap of such plastic is not reusable. Bakelite is an example of such plastic.

2. **Thermoplastic:** In this variety, the linkage between the molecules is very loose. They can be softened by heating repeatedly. This property helps for reuse of waste plastic. These plastic need time to cool down and harden. These plastics are to be kept in moulds till cooling takes place completely. Bitumen, cellulose and shellac are the examples of this variety of plastics.

Properties of Plastics

1. **Colour:** Some plastics are completely transparent. Using pigments plastics of any attractive colour can be produced.

2. **Dimensional Stability:** It is dimensionally stable to a great extent.
3. **Durability:** Plastic offers great resistance to moisture and chemicals and hence more durable.
4. **Electrical Insulation:** The plastics possess excellent electrical insulating property.
5. **Fire Resistance:** The phenol-formaldehyde and urea-formaldehyde plastics resist fire to a great extent and hence they are used as fire proofing materials.
6. **Strength:** The plastics are reasonably strong. Their strength may be increased by reinforcing with various fibrous materials. Attempts are being made to produce structurally sound plastics.
7. **Specific Gravity:** The specific gravity of plastics is very low and hence convenient to handle.
8. **Ductility:** The plastics are not ductile and hence they fail without giving warning.
- 9 **Maintenance:** There is no maintenance cost for plastic articles *i.e.*, they do not need painting and polishing.

Uses of Plastics

There are variety of plastics made to suit different uses. The typical uses of plastics in buildings is listed below:

1. Corrugated and plain sheets for roofing.
2. For making joint less flooring.
3. Flooring tiles.
4. Overhead water tanks.
5. Bath and sink units.
6. Cistern hall floats.
7. Decorative laminates and mouldings.
8. Window and door frames and shutters for bathroom doors.
9. Lighting fixtures.
10. Electrical conduits.
11. Electrical insulators.
12. Pipes to carry cold waters.

GLASS

Silica is the main constituent of glass. But it is to be added with sodium potassium carbonate to bring down melting point. To make it durable lime or lead oxide is also added. Manganese oxide is added to nullify the adverse effects of unwanted iron present in the impure silica.

Types of Glass

The glass may be broadly classified as:

1. Soda-lime glass 2. Potash lime glass
3. Potash lead glass 4. Common glass and
5. Special glasses.

1. **Soda Lime Glass:** It is mainly a mixture of sodium silicate and calcium silicate. It is fusible at low temperature. In the fusion condition it can be blown or welded easily. It is colourless. It is used as window panes and for the laboratory tubes and apparatus.

2. **Potash Lime Glass:** It is mainly a mixture of potassium silicate and calcium silicate. It is also known as hard glass. It fuses at high temperature. It is used in the manufacture of glass articles which have to withstand high temperatures.

3. **Potash Lead Glass:** It is mainly a mixture of potassium silicate and lead silicate. It possesses bright lustre and great refractive power. It is used in the manufacture of artificial gems, electric bulbs, lenses, prisms etc.

4. **Common Glass:** It is mainly a mixture of sodium silicate, calcium silicate and iron silicate. It is brown, green or yellow in colour. It is mainly used in the manufacture of medicine bottles.

5. **Special Glasses:** Properties of glasses can be suitably altered by changing basic ingredients and adding few more ingredients. It has now emerged as versatile material to meet many special requirement in engineering. The following is the list of some of the special glasses:

- (a) Fibre glass (b) Foam glass
- (c) Bullet proof glass (d) Structural glass
- (e) Glass black (f) Wired glass
- (g) Ultraviolet ray glass (h) Perforated glass.

Important Properties of Glass

1. It absorbs, refracts or transmits light. It can be made transparent or translucent.
2. It can take excellent polish.
3. It is an excellent electrical insulator.
4. It is strong and brittle.
5. It can be blown, drawn or pressed.
6. It is not affected by atmosphere.

7. It has excellent resistance to chemicals.
8. It is available in various beautiful colours.
9. Glass panes can be cleaned easily.

TILES

These are also clay products like brick but are thin. Depending upon their use, building tiles may be further classified as

1. Roofing tiles
2. Flooring tiles and wall tiles.

1. **Roofing Tiles:** Roofing tiles are used to cover sloping roofs. They are supported on wooden reapers. Sometimes light gauge steel or steel rods are also used as reapers. After supporting on reapers these tiles should be strong enough to take load of a man safely. The tiles should be leak proof. Normally these tiles are having curved surface having ribbed sections, so that with thin section they are sufficiently strong to resist the load. These tiles are not subjected to load directly. They serve in reducing adverse thermal effects. Mangalore, Allahabad tiles, and corrugated tiles are popularly used roofing tiles.

FIG-INCOMPLETE

The **desirable properties** of the roofing tiles are:

1. They should not absorb moisture more than 20 per cent by weight.
2. They should give pleasing look.
3. They should be capable of taking load of a man safely, after they are supported on reapers.
4. They should be durable.
5. They should be uniform in shape and size.

2. **Flooring Tiles and Wall Tiles:** These tiles are manufactured by burning pressed green tiles twice. First they are burnt at 700°C. Then they are dipped in the glaze solution and again burnt at 1250°C to fuse them with glaze. The thickness of these tiles vary from 15 to 20 mm. These tiles are flat and they have pleasing appearance. There are two types of flooring tiles:

(a) **Glazed Tiles:** These tiles are used as finish surfaces for floors and walls in kitchen and bathrooms. These tiles are glazed and are provided with attractive colours and designs.

(b) **Mosaic Tiles:** These are precast concrete tiles with marble chips on the top surface. After fixing these tiles polishing is done.

The desirable properties of flooring and roofing tiles are:

1. Tolerance for length = ± 5 mm.

2. Tolerance for thickness = ± 2 mm.
3. Should be uniform in shape and colour.
4. They should be sound, hard and durable.
5. They should have very low percentage of water absorption.
6. They should give a clear ringing sound when struck with each other.
7. They should show good resistance to abrasion.

ADHESIVES

An adhesive is a substance which is used to join 2 or more parts so as to form a single unit.

The application of adhesives has the following advantages.

1. A wide variety of combination in joining is possible.
2. It can be used for bonding the surfaces of glass, metal, plastic and wood.
3. It creates a massive effect.
4. It is possible to prevent corrosion between different metals joined by adhesives.
5. It produces adequate strength.
6. The process of applying adhesives is easy, economical and speedy.

The disadvantages of adhesives are as follows

1. It is not possible to adopt any adhesives for all substances. Depending upon the properties of the substances to be joined, suitable adhesive has to be selected.
2. The adhesive substance does not become strong immediately after its application. It requires some time to attain the desired strength.
3. The adhesive substance generally does not remain stable at high temperature.

The glue is a general term which is used to indicate an adhesive substance. Following are the various types of adhesives.

1. **Albumin Glues:** It is a glue of better quality. It is not attacked by water. It is used for making furniture.
2. **Animal protein glues:** It is obtained by boiling waste pieces of skins, bones, etc. of animals with hot water. The animal glue develops strong and tough joints and it is easy to apply. But it is affected by damp and moist conditions.
3. **Glues from natural resins:** It is prepared from natural resins. It is used for labelling building paper etc.
4. **Glues from synthetic resins:** These glues are based on synthetic resins. They may be either be thermos-setting glues or thermos-plastic glues.

5.Nitrocellulose glues: It is prepared from pyroxilin which is nitrated cellulose. It is derived by treating cellulose with nitric acid.

6.Rubber glues: It is prepared by dissolving rubber in benzene. It is used for joining rubber, plastics, glass etc.

7.Special glues: These are specially prepared to join metals.

8.Starch glues: It is prepared from vegetable starch. It has good strength in dry condition. But it is not moisture resistant.

9.Vegetable starch: It is prepared from natural gums and starches. It is used for preparing paper board articles, labelling etc.

COMPOSITE MATERIALS

The composite materials are shortened as composites. They are formed by combining two or more different materials to make better use of their virtues and by minimizing their deficiencies.

The composites are made from two main constituent materials:

1. strong load carrying materials known as reinforcement or reinforcing fibres:
2. weaker material known as matrix.

Reinforcing fibres: Following are the functions of reinforcing fibres:

- 1.It provides strength and rigidity.
- 2.It helps to support structural load.

There are three most common types of reinforcing fibres.

- 1.Glass Fibre
- 2.Carbon
- 3.Aramid

Matrix: Following are the functions of matrix:

- 1.It works as a binder.
- 2.It maintains the position and orientation of the reinforcement.
- 3.It balances loads between the reinforcement.
- 4.It protects the reinforcement degradation. It provides shape and form to the structure.

The most common type of matrix is thermosetting résins.

Following are the common examples of man-made composites.

Wattle and daub (mud and straw) is one of the most ancient man-made composite.

Bricks made from straw and mud.

Plywood is the building block of a composite laminate. It is a thin layer of composite material in which fibre is usually oriented in one direction.

Concrete is a classic example of a ceramic composite, which consist of the particles of sand and aggregates of graded sizes in a matrix of hydrated Portland cement.

STEEL

As far as the carbon content is concerned, the steel forms an intermediate between cast-iron and wrought-iron. The cast-iron contains carbon from 2 to 4 per cent. In wrought-iron, the carbon content does not exceed 0.15 per cent steel, the carbon content varies from anything below 0.25 per cent to 1.50 per Cent maximum.

The steel becomes harder and tougher as its carbon content goes on increasing and at the maximum level of 1.5 per cent, all the carbon gets into chemical combination with iron and none of it exists in its free state.

The steel is suitable for all constructional purposes in general and hence it has practically replaced cast-iron and wrought-iron in the present day practice of building construction. It is equally strong in compression as well as in tension.

Uses of Steel

Depending upon the carbon content, the steel is designated as mild steel or medium carbon steel or high carbon steel.

- The carbon content of mild steel is about .010 to .25 %. When carbon content is less than .10% it is known as dead steel and very low carbon steel.
- The carbon content of medium carbon steel is about .25 to .60%.
- The high carbon steel is also known as the hard steel and its carbon content varies from .60 to 1.10%.

Uses of steel

Name of steel	Carbon content	Uses
Mild steel	up to .10%	Motor body, sheet metal, tin plate etc.
Medium carbon steel	up to .25%	Boiler plates, structural steel etc.
	up to .45%	Rails, tyres etc.
	Up to .60%	Hammers, large stamping and pressing dies etc.
High carbon steel	up to .75%	Sledge hammers, springs, stamping dies
	up to .90%	

	up to 1.00%	Miner's drills, smith tools, stone mason's tools etc.
	up to 1.10%	Chisels, hammers, saws, wood working tools Axes, cutlery, drills, knives, picks, punches

Magnetic properties of Steel

The steel is also widely used in electrical machinery, generators, transformers, etc. For making steel suitable for such use, its magnetic properties are given supreme importance and these properties are obtained by carefully adjusting its chemical composition. Following are the proportions of various elements in steel for making it to achieve better magnetic properties:

- (1) **Carbon:** It is desirable to keep carbon content as low as possible and it should not exceed 0.10 per cent.
- (2) **Silicon:** The presence of silicon results in considerable increase of electrical losses and hence it is highly undesirable.
- (3) **Sulphur and phosphorus:** If combined content of sulphur and phosphorus exceeds about 0.30 per cent, the magnetic properties of steel are greatly affected.
- (4) **Manganese:** If content of manganese exceeds about 0.30 per cent, it proves to be injurious to the magnetic properties of steel.

Properties of mild steel

Following are the properties of mild steel:

- It can be magnetised permanently.
- It can be readily forged and welded
- It cannot be easily hardened and tempered.
- It is not easily attacked by salt water.
- It is used for all types of structural work
- It rusts easily and rapidly.
- Its melting point is about 1400°C.
- Its specific gravity is 7.80.
- Its ultimate compressive strength is about 80 to 120 kN per cm².
- Its ultimate tensile and shear strengths are about 60 to 80 kN per cm²

Properties of hard steel

Following are the properties of hard steel:

- It can be easily hardened and tempered.
- It can be magnetised permanently
- It cannot be readily forged and welded.
- It has granular structure.
- It is not easily attacked by salt water
- It is tougher and more elastic than mild steel
- It rusts easily and rapidly
- Its melting point is about 1300°C.
- Its specific gravity is 7.90.
- Its ultimate compressive strength is about 140 to 200 kN per cm²,
- Its ultimate shear strength is about 110 kN per cm². Its ultimate tensile strength is about 80 to 110 kN per cm²

Market forms of Steel

Following are the standard shapes in which the rolled steel sections are available in the market:

- 1.Angle sections
- 2.Channel sections
3. Flat bars
- 4.Corrugated sheets
- 5.Expanded metal
- 6.I-sections
- 7.Plates
- 8.Ribbed (HYSD) bars
- 9.Square bars
- 10.Ribbed mild steel bars
- 11.Thermo-Mechanically Treated bars (TMT bars)
- 12.Cold Twisted Deformed bars (CTD bars)
- 13.Welded Wire Fabrics (WWF).

NON-FERROUS METALS

Following Eight non-ferrous metals which have limited use in the engineering structures will be studied in this chapter:

- | | | | |
|-------------|-----------|--------------|----------|
| 1. Aluminum | 3. Copper | 5. Magnesium | 7. Tin |
| 2. Cobalt | 4. Lead | 6. Nickel | 8. Zinc. |

ALUMINIUM

The aluminium occurs in abundance on the surface of earth. It is available in various forms such as oxides, sulphates, silicates, phosphates, etc. But it is commercially produced mainly from bauxite ($\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$) which is hydrated oxide of aluminium.

Properties of Aluminium

Following are the properties of aluminium:

- It is a very good conductor of heat and electricity
- It is a silvery white metal with bluish tinge and it exhibits bright lustre on a freshly broken surface.
- It is a non-magnetic substance.
- It is light in weight, malleable and ductile.
- It is very soft.
- Its specific gravity is about 2.70.
- It melts at 660°C and its boiling point is 2056°C .
- It possesses great toughness and tensile strength.
- It readily dissolves in hydrochloric acid.

Uses of Aluminium

This metal is chiefly used for making parts of aeroplane, cooking utensils, electric wires, window frames, glazing bars, corrugated sheets, structural members, tools, posts, panels, balustrades, bathroom fittings, precision surveying instruments, furniture, etc. Its other uses can be mentioned as follows:

- It is used as a reducing agent in the manufacture of steel.
- It is used for making aluminium alloys, automobile bodies, engine parts and surgical instruments.
- It is used in the casting of steel.

- It is used in the manufacture of electrical conductors
- It is used in the manufacture of paints in powder form.

COBALT

The cobalt is found to occur in free state in meteorites. Its two important ores are arsenide and sulphoarsenide.

Properties of Cobalt

Following are the properties of cobalt:

- If cobalt is red hot, it can decompose steam.
- It is a lustrous white metal
- It is malleable and ductile.
- It is not affected by atmosphere at ordinary temperature.
- It is not attacked by alkalis
- Its specific gravity is 8.90.
- It melts at 1480°C and its boiling point is 2900°C.
- It is magnetic and can retain its magnetic properties up to a temperature of about 1100°C.

Uses of Cobalt

The cobalt is widely used in the preparation of special alloy steels, ceramic products, television articles, etc. It forms the basis of all blue colours used in glass and porcelain manufacture.

COPPER

The copper occurs in practically all important countries of the world. Its principal ores are Cuprite Cu_2O , Copper glance Cu_2S , Copper pyrites CuFeS_2 , Malachite $\text{CuCO}_3 \cdot \text{Cu(OH)}_2$, and Azurite $2\text{CuCO}_3 \cdot \text{Cu(OH)}_2$.

Properties of Copper

Following are the properties of copper:

- It becomes brittle just below its melting point.
- It can be worked in hot or cold condition, but it cannot be welded
- It is a good conductor of heat and electricity
- It is attacked by steam at white heat.
- It is malleable, ductile and extremely soft.
- It melts at 1083°C and its boiling point is 2300°C.
- Its specific gravity is 8.92.

Uses of Copper

The market forms of copper are ingots, sheets, tubes and wires. It is extensively used for making electric cables, alloys, household utensils, electroplating, lighting conductors, dowels in stone masonry, blocks for printing, etc. It is mainly used in the manufacture of alloys of which brass and bronze are important.

LEAD

The lead occurs occasionally in free state in nature. In combined form, it mainly occurs as sulphide, the ore being known as the Galena PbS. This is the most important and widely distributed ore of lead. It contains about 86 per cent lead and 14 per cent sulphur. It is always associated with silver sulphide and often gold. A pure galena possesses a metallic bluish lustre of dark grey colour.

Properties of lead

Following are the properties of lead:

- It can be cut with a knife
- It makes impression on paper.
- It melts at 327.5°C and its boiling point is 1620°C.
- It is not attacked by dry air, but moist air takes away its lustre bright metallic and a dark protective film is produced on the surface of metal.
- It is readily dissolved in dilute nitric acid.
- It is very soft, plastic and almost devoid of elasticity.
- Its specific gravity is 11.36.

Uses of lead

The lead is widely used for making shots, bullets, alloys, storage cells, sanitary fittings, cisterns, water-proof and acid-proof chambers, gas pipes, roof gutters, printing types, damp-proof courses of buildings, cable coverings, preparation of lead oxides for paints, etc.

MAGNESIUM

The magnesium does not occur in free state in nature, but it is a widely distributed element found in the combined state. Its principal ores are magnesite MgCO_3 , dolomite CaCO_3 , MgCO_3 , kieserite MgSO_4 , H_2O and carnallite $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$.

Properties of Magnesium

Following are the properties of magnesium:

- It burns when heated in air with a dazzling bluish-white light extremely rich in ultra-violet rays.

- It carries away heat easily.
- If strongly heated, it can decompose steam. If it is in the form of finely divided particles, it burns readily and easily.
- It is a silver-white metal possessing a high lustre.
- It is ductile and malleable.
- It is not affected by alkalis.
- Its melting point is 651°C and it boils at 1110°C .
- Its thermal coefficient of expansion is high.

Uses of magnesium

The magnesium is used in photography, fire-works, signalling, paints, whitening paper pulp, etc. It alone cannot be used for structural work. But some of its alloys may be employed for some structural parts. It is used as a refractory material in the form of magnesia bricks for the lining of steel and electric furnaces. It is also used as a cementing material for the manufacture of artificial stone, plaster, tiles, etc.

Nickel

The nickel occurs in free state in meteorites. In combination, it chiefly occurs as sulphide ores and silicate ores. The most important nickel ore is a mineral composed of magnetic iron pyrites with nickel.

Properties of Nickel

Following are the properties of nickel:

- If nickel is red hot, it can decompose steam.
- If it is in a finely ground powder form, it may absorb hydrogen to the extent of about 17 times its volume.
- It is a greyish white lustrous metal.
- It is capable of taking a high polish and can easily be welded.
- It is hard, malleable and magnetic.
- It is not attacked by fused alkalis.
- Its resistance to corrosion is high.
- Its specific gravity is 8.90 and melts at 1452°C . Its boiling point is 2900°C .

Uses of nickel

The nickel is widely used as a coating for other metals and for the preparation of alloys like German silver, nickel steels, etc. Its practical use is confined to the manufacture of chemical apparatus, crucible, electroplating parts of machines, domestic utensils, etc.

TIN

The tin occurs chiefly as tinstone or cassiterite which is its oxide, SnO_2 . It is also available in nodules which are known as the stream tin.

Properties of Tin

Following are the properties of tin:

- It becomes brittle when heated to a temperature of about 200°C .
- It melts at 231.5°C and boils at 2260°C .
- It is a white metal with a brilliant lustre.
- It is not affected by dry air.
- It is not attacked by pure water.
- Its specific gravity is 7.31
- It is soft and malleable.
- It withstands corrosion due to acids

Uses of tin

The tin is rarely used alone. It is used for plating, lining lead pipes and for the preparation of alloys and solder. The pure tin is commercially used for making evaporating basins, infusion pots, etc. It is also used for giving a protective coating to the copper and iron utensils. The steel sheets coated with tin are used for making cans for food, fruit and milk industry. The tin-foils are used for silvering of mirrors and for wrapping up cheese, chocolates, tobacco, toilet soaps, etc.

ZINC

The zinc does not occur in free state in nature. Its principal ores are zincite or red zinc ZnO , franklinite ZnO , Fe_2O_3 , calamine or zinc-copper ZnCO_3 and zinc blende ZnS . A good quality of zinc blende may contain 50% of zinc. The pure ZnS contains about 65% of zinc. It is the chief source of the metal.

Properties of Zinc

Following are the properties of zinc:

- It burns with a greenish white flame when strongly heated in air.
- It may be drawn into wires and rolled into sheets between temperature range of 100°C to 150°C ,
- It melts at 419.4°C and boils at 907°C .
- It is a bluish white metal.
- It is brittle at the ordinary temperature.

- It is a good conductor of heat and electricity.
- It is not affected by dry air.
- It is not affected by pure water
- It resists corrosion.
- Its specific gravity is 7.14.
- It should be kept clear of lime and calcareous substances.
- It is harmfully attacked and ultimately destroyed by acid, hot water and Sea salt.

Uses of Zinc

The zinc is used in electric cells, for galvanizing, in the preparation of alloys, paints, etc. The zinc has great value as a protective covering or coat to iron works , plain and corrugated sheets, iron vessels, etc. and as a lining to the drinking water storage tanks.